

Polynomials – 8

Several variables and the polynomial method

Homework

8 сентября 2026 года

Theoretical minimum. The grid lemma; the Combinatorial Nullstellensatz and its coefficient hypothesis; the Schwartz–Zippel bound; homogeneous components; the difference between a polynomial and the function it defines; Chevalley–Warning as a special case of the method, proved in Polynomials – 6.

1. Prove that a non-zero polynomial in n variables over an infinite field is not the zero function, and give a non-zero polynomial over $\mathbb{F}_p$ in one variable which vanishes everywhere.
2. (Putnam 2005) Find a non-zero polynomial $P(x, y)$ such that $P(\lfloor a \rfloor, \lfloor 2a \rfloor) = 0$ for all real numbers a .
3. (Proposed problems) For positive integers n and k determine the dimension of the linear space of polynomials in n variables over a field K of degree at most k , and deduce the number of monomials of degree exactly k .
4. (Putnam 1986) Let $f(x, y, z) = x^2 + y^2 + z^2 + xyz$ and let p, q, r be real polynomials with $f(p, q, r) = f$. Prove or disprove that p, q, r is a permutation of $\pm x, \pm y, \pm z$ in which the number of minus signs is 0 or 2.
5. (IMS 2022) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a polynomial whose degree in each variable is at most one, and let $Q = [0, 1]^n$. Show that there is a point b with coordinates 0 or 1 such that $|f(b)| \geq |f(x)|$ for every $x \in Q$.
6. (Высшая лига 2012) Find at least one polynomial $u(x, y)$ with $u_x + 2xu_y = 0$, $u(0, 0) = 1$ and $u(0, 1) = u(0, 2) = u(0, 3) = 0$.

7. (Mexmat, 2017) Prove that there is a non-zero homogeneous polynomial $f(x, y, z)$ such that $f(a, b, c) = 0$ whenever a, b, c are the sides of a triangle in which the angle opposite a is 2017 times the angle opposite b .
8. Prove that the Schwartz–Zippel bound is sharp for every degree, and deduce a randomised test for the identity of two multivariate polynomials given as black boxes.
9. Prove Alon’s non-vanishing criterion from the grid lemma, and use it to show that for a prime p any $2p - 1$ integers contain p whose sum is divisible by p .
10. (MIPT) Let $P(z_1, z_2)$ be a homogeneous polynomial of degree n in two complex variables. Prove that every point at which $|P|$ attains its maximum on the unit sphere $S^3 \subset \mathbb{C}^2$ is at distance at least $\arcsin \frac{1}{n}$, in the inner metric of the sphere, from the intersection of the zero set of P with S^3 .